

JOBSHIELD - AI-POWERED JOB SCAM DETECTOR

FULL ARCHITECTURE & SYSTEM DESIGN REPORT

1. EXECUTIVE SUMMARY & HOW IT WORKS

JobShield combines Google Gemini 3.6 Flash, Apache PDFBox document parsing, and Jaccard keyword pattern matching to protect job seekers from scam offers.

- Step 1: User submits text or uploads offer letter PDF.
- Step 2: Spring Boot backend validates JWT authentication and extracts text.
- Step 3: Google Gemini AI evaluates scam indicators and returns Risk Score.
- Step 4: ScamSimilarityEngine matches patterns (FEE, WHATSAPP, TELEGRAM).
- Step 5: Postings $\geq 80\%$ similarity are grouped into ScamCampaign clusters.
- Step 6: MySQL stores analysis; frontend updates dashboard & history.

2. TECHNOLOGY STACK

- Frontend: React 19, Vite, React Router, Pure CSS (Cybersecurity Dark Theme)
- Backend: Java 17, Spring Boot 4.x / 3.x, Spring Security, JWT (JJWT 0.12.7)
- AI Engine: Google Gemini 3.6 Flash via Spring RestClient
- Document Engine: Apache PDFBox 3.0.3 (text extraction from PDFs)
- Database: MySQL (jobshield_db) with Hibernate JPA ORM

3. REACT COMPONENT ARCHITECTURE

- App.jsx: Main routing tree with public and ProtectedRoute gatekeepers.
- api.js: Centralized fetch service with automatic JWT injection & error handling.
- ProtectedRoute.jsx: Guards private pages and renders persistent Sidebar.
- AnalysisResultCard.jsx: Visual score card (0-100), AI explanation, and badges.
- DashboardPage.jsx: Live threat metrics and pure CSS risk distribution bar.
- AnalyzeJobPage.jsx: Text analysis form connected to POST /api/jobs/analyze.
- PdfScannerPage.jsx: Drag-and-drop PDF upload connected to /api/jobs/analyze-pdf.
- HistoryPage.jsx: User scan log with modal drilldown (/api/jobs/{id}).
- CampaignsPage.jsx: Multi-company scam syndicate intelligence.

4. SAMPLE TEST PDFS PROVIDED

- sample_pdfs/Sample_Scam_Job_Offer_Letter.pdf (Risk Score: 100 - HIGH)
- sample_pdfs/Sample_Legitimate_Job_Offer_Letter.pdf (Risk Score: 10 - LOW)

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